



Cambridge International AS & A Level

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MATHEMATICS

9709/21

Paper 2 Pure Mathematics 2

May/June 2026
1 hour 15 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

 This document has **12** pages.

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- 2 (a) Solve the equation $|3x + 7| = |4x - 1|$. [3]

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- (b) Hence, solve the equation $|3 \cot \theta + 7| = |4 \cot \theta - 1|$ for $0 < \theta < \pi$. Give each solution correct to 3 decimal places. [3]

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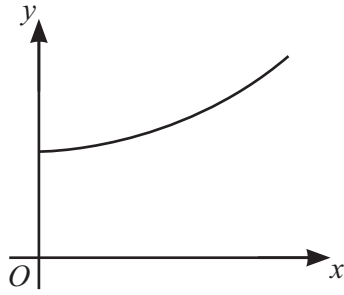
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3



The diagram shows the curve with equation $y = \sqrt{x^3 + 9}$ for $0 \leq x \leq 3$.

- (a) Use the trapezium rule with three intervals to find an approximate value of $\int_0^3 \sqrt{x^3 + 9} \, dx$.

Give your answer correct to 3 significant figures.

[3]

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- (b) State, with a reason, whether your answer to **3(a)** is an over-estimate or under-estimate of the true value of $\int_0^3 \sqrt{x^3 + 9} \, dx$.

[1]

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- 4 (a) Express $5 \cos \theta + 3 \sin \theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give the exact value of R , and give the value of α correct to 2 decimal places. [3]

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- (b) Hence, solve the equation $5 \cos \theta + 3 \sin \theta = 2$ for $-180^\circ < \theta < 180^\circ$. [4]

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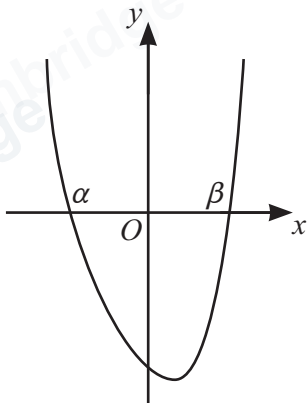
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$$p(x) = 2x^4 + 3x^3 + 4x^2 - 8x.$$

- [illegible]



(b) State the value of α , and show that β satisfies the equation $x = \sqrt[3]{7-2x}$. [2]

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- (c) Show by calculation that the value of β lies between 1.5 and 1.6. [2]

- (d) Use an iterative formula, based on the equation in 5(b), to find the value of β correct to 4 significant figures. Give the result of each iteration to 6 significant figures. [3]



$$x = 1 + 2e^{2t}, \quad y = \frac{e^t}{9 + e^{2t}}.$$

(a) Find an expression for $\frac{dy}{dx}$ in terms of t .

[4]

[illegible]



(b) Find the coordinates of M .

[3]

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(c) Find the gradient of the curve at the point for which $x = 33$.

[3]

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$$y = 3 + \sin^2 2x - \cos 2x$$

Find the exact area of the shaded region.

[9]

This image shows a full page of blank handwriting practice paper. It features multiple sets of horizontal lines. Each set consists of a solid top line, a dashed middle line, and a solid bottom line, providing a guide for letter height and placement. The background is white, and the lines are light gray or blue. There are no margins, text, or other markings on the page.

[illegible]

[illegible]

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